

Theoretical Foundations of Option Pricing Models

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Abstract

This document provides a detailed overview of the mathematical models implemented in the accompanying Python script (`option_pricer.py`). It covers the assumptions, formulas, parameters, and limitations of the Black-Scholes and Black '76 models for pricing various types of financial options.

1 Core Assumptions of the Black-Scholes World

The foundational models described here operate under a specific set of idealized market conditions known as the "Black-Scholes world." These assumptions are crucial for the derivation of the formulas.

- **Random Walk:** The stock price follows a Geometric Brownian Motion, meaning price changes are random and unpredictable.
- **Constant Volatility and Interest Rates:** The volatility (σ) of the underlying asset and the risk-free interest rate (r) are known and constant over the life of the option.
- **No Dividends (Original Model):** The original model assumes the underlying stock does not pay dividends. Extensions (like Merton's model) adjust for this.
- **Frictionless Market:** There are no transaction costs (e.g., commissions or brokerage fees) or taxes. Securities are perfectly divisible.
- **Efficient Markets:** All relevant information is immediately reflected in the stock price. Arbitrage opportunities do not exist.
- **Lognormal Distribution:** Stock prices are lognormally distributed, which means that the logarithm of the stock price is normally distributed.

2 Black-Scholes Model (1973) for European Options

This is the cornerstone model for pricing European options, which can only be exercised at expiration.

2.1 Formulas

The price of a European call option (C) and a European put option (P) are given by:

$$C(S, t) = Se^{-q(T-t)}N(d_1) - Ke^{-r(T-t)}N(d_2)$$

$$P(S, t) = Ke^{-r(T-t)}N(-d_2) - Se^{-q(T-t)}N(-d_1)$$

where:

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$
$$d_2 = d_1 - \sigma\sqrt{T - t}$$

2.2 Parameters Used

Parameter	Value
S (Current Stock Price)	100
K (Strike Price)	100
T (Time to Maturity)	1.0 year
r (Risk-free Rate)	0.05 (5%)
σ (Volatility)	0.20 (20%)
q (Dividend Yield)	0 (for base case)
$N(\cdot)$	CDF of Standard Normal Dist.

2.3 Numerical Results

Option Type	Calculated Price
European Call	\$10.4506
European Put	\$5.5704

Note: The calculated price represents the premium an investor pays per share for one option contract. For the call option, this is the cost today (\$10.4506 per option contract) for the right, but not the obligation, to purchase the stock for \$100 at expiration.

2.4 Limitations

The model's assumptions are its primary limitation. In reality, volatility and interest rates are not constant, markets are not frictionless, and stock price movements do not perfectly follow a lognormal distribution (e.g., "fat tails").

3 Black '76 Model for Options on Futures

This is an adaptation of the Black-Scholes model for pricing European options on futures contracts. The key difference is that futures contracts do not require an initial investment, which simplifies the cost-of-carry component.

3.1 Formulas

$$C(F, t) = e^{-r(T-t)}[FN(d_1) - KN(d_2)]$$

$$P(F, t) = e^{-r(T-t)}[KN(-d_2) - FN(-d_1)]$$

where d_1 and d_2 are calculated using the futures price F instead of the spot price S , and the cost-of-carry is assumed to be zero.

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

3.2 Parameters Used (At-the-Money)

Parameter	Value
F (Current Futures Price)	100
K (Strike Price)	100
T (Time to Maturity)	1.0 year
r (Risk-free Rate)	0.05 (5%)
σ (Volatility)	0.20 (20%)

3.3 Numerical Results (At-the-Money)

Option Type	Calculated Price
European Call on Future	\$7.9656
European Put on Future	\$7.9656

Note: The calculated price is the premium for one option contract on a future. A buyer pays this amount (\$7.9656 per option contract) for the right to enter into the underlying futures contract at the strike price at expiration. The call and put prices are identical because the option is "at-the-money" (Futures Price = Strike Price). This is a direct result of the put-call parity relationship for futures options: $C - P = (F - K)e^{-rT}$. Since $F = K$, the right side of the equation becomes zero, forcing $C = P$.

3.4 Limitations

The model shares the same core limitations as Black-Scholes concerning constant volatility and interest rates. It is also only applicable to European-style options.

4 Examples

4.1 Black '76 Model: Out-of-the-Money vs. In-the-Money

This example shows how the Black '76 prices change when the option is no longer at-the-money. We change the Strike Price to \$110, while the Futures Price remains \$100.

4.1.1 Parameters Used (Out-of-the-Money Call / In-the-Money Put)

Parameter	Value
F (Current Futures Price)	100
K (Strike Price)	110
T (Time to Maturity)	1.0 year
r (Risk-free Rate)	0.05 (5%)
σ (Volatility)	0.20 (20%)

4.1.2 Numerical Results

Option Type	Calculated Price
European Call on Future	\$4.2134
European Put on Future	\$13.7257

Note: The call option is now "out-of-the-money" as the strike (\$110) is above the futures price (\$100). Its premium falls because the market has to rally significantly for it to be profitable. Conversely, the put option is "in-the-money" as it gives the right to sell at \$110, which is higher than the current price. Its premium increases substantially. The price difference reflects the discounted value of the \$10 difference between the strike and futures price, as dictated by put-call parity.